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FINAL RESEARCH REPORT

Spatiotemporal Video Processing Lab: Video Stabilization, Background/Foreground Separation, and Comparative DSP Algorithm Evaluation

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1. Abstract

Camera shake corrupts the pixel-wise background models used to separate moving objects from a scene, yet stabilization and background subtraction are rarely evaluated together under identical conditions. This report documents the SpatioTemporal Video Processing Lab, a browser-native workbench (React, TypeScript, OpenCV.js WebAssembly) that implements three stabilizers (no stabilization, pyramidal Lucas–Kanade with 2D similarity RANSAC, and Enhanced Correlation Coefficient alignment), three background models (Frame Differencing, MOG2 and a project Sample-based KNN), a morphological post-processing chain, a combined pipeline and an evaluation framework that distinguishes ground-truth, proxy and runtime metrics. We assemble a licensed dataset of 15 short videos (five stabilization, five background subtraction, five combined) and run the project's experiment groups using a Python/OpenCV reference implementation that follows the application's default parameters, then the application's own Benchmark Lab on every video. With known synthetic camera motion, both stabilizers remove more than 90 % of the injected jitter at the Medium and High levels (High: 3.68 px → 0.14 px). On real video, LK + RANSAC reduces re-measured jitter by 86 to 89 % on mild-to-moderate shake but loses most of its benefit on the most severe clips, where the RANSAC inlier ratio is low, while ECC is less robust to large displacement. MOG2 is the most stable background model under illumination drift (0.4 % versus about 19 % foreground on a clouds time-lapse). Ablation shows that shadow suppression and morphology influence the foreground mask most, and that removing RANSAC doubles residual jitter. Stabilizing before background subtraction yields only small, proxy-level differences. The application's own jitter metric, computed on the trajectory it estimates, reports 81 to 97 % reduction for LK + RANSAC on all 15 videos, including clips where the independent re-measurement shows little benefit, so it should be read as a smoothness measure rather than proof of correctness. Accuracy claims are withheld because real videos have no ground-truth masks.

2. Keywords

Video Stabilization, Background Subtraction, Foreground Segmentation, Lucas–Kanade Optical Flow, RANSAC, Enhanced Correlation Coefficient (ECC), Gaussian Mixture Model (MOG2), Sample-based KNN, Frame Differencing, Morphological Filtering, Ablation Study, OpenCV.js WebAssembly, Digital Signal Processing (DSP).

3. Introduction

Video is a spatiotemporal signal: every pixel is a one-dimensional time series, and every frame is a two-dimensional sampled field. Two classical operations on this signal underpin many surveillance, robotics and scientific-imaging applications. **Video stabilization** estimates the unwanted high-frequency component of camera motion and removes it by geometrically warping each frame. **Background/foreground separation** models the temporally stable part of a scene and labels the remaining pixels as moving foreground. The two operations interact: a background model assumes that a given pixel keeps observing the same scene point, an assumption that camera shake violates, so shaking frames produce false foreground along every textured edge.

Most published work evaluates stabilization and background subtraction separately, and most software that implements them is either a closed tool or a collection of scripts that cannot be inspected stage by stage. This project therefore builds a single browser-native experimental workbench, the **SpatioTemporal Video Processing Lab**, in which every frame is decoded, processed and quantitatively evaluated on the client with OpenCV.js (WebAssembly) and TypeScript numerical routines. The goal is not to invent a new algorithm but to provide a unified, reproducible environment for comparing the role, effectiveness and limitations of classical DSP algorithms.

This report documents the system as implemented in the project source code, describes the 15-video dataset assembled for it, and reports an experimental study organized as in the project's research design: a stabilization comparison (A0/A1/A2), a background-subtraction comparison (B1/B2/B3), a combined-pipeline study (Flow A versus Flow B), a seven-variant ablation (D0–D6) and a one-parameter-at-a-time sensitivity analysis.

4. Research Problem

How do different classical stabilization algorithms (sparse Lucas–Kanade tracking with RANSAC versus dense ECC alignment) differ in the stability and computational cost they deliver, does stabilizing a shaky video before background subtraction measurably change the foreground masks, and how does removing or reordering individual DSP stages change output quality and runtime?

5. Research Objectives

The primary objectives of this research project are:

- **Objective 1 (Unified Workbench):** Implement a web application that performs video stabilization, background/foreground separation and their combination on user-supplied video, with a decoder that enforces resource limits (90 frames, 480 px width, 15 fps and 12 s by default).
- **Objective 2 (Algorithm Comparison):** Implement and compare two stabilizers (A1 LK + RANSAC, A2 ECC) against an unstabilized baseline (A0), and three background models (B1 Frame Differencing, B2 MOG2, B3 Sample-based KNN) with identical morphological post-processing.
- **Objective 3 (Evaluation Framework):** Provide quantitative metrics that separate ground-truth metrics (IoU, F1, Transform RMSE) from proxy metrics (Jitter RMS, foreground ratio, mask flicker) and runtime metrics, and refuse to report accuracy when no ground truth exists.
- **Objective 4 (Ablation and Sensitivity):** Quantify the contribution of each DSP stage by single-stage ablation and the influence of key parameters by one-at-a-time sweeps.
- **Objective 5 (Dataset):** Assemble a small, licensed, reproducible video dataset with three categories (stabilization, background subtraction, combined) of five videos each.

6. Research Questions & Hypotheses

Following the project design document, three research questions and seven hypotheses are formulated:

- **RQ1 (Stabilization):** How do LK + RANSAC and ECC differ in video stability and processing speed?
- **RQ2 (Background Subtraction):** How effective are Frame Differencing and MOG2 under a fixed camera, camera shake, dynamic background and illumination change?
- **RQ3 (Processing Order and Ablation):** How does the order of stabilization and background subtraction affect foreground quality, and how does removing each DSP stage change quality and performance?

Formal Research Hypotheses:

- **H1:** Trajectory smoothing reduces unwanted camera-trajectory oscillation.
- **H2:** LK + RANSAC and ECC perform differently depending on the camera-motion type and the video content.
- **H3:** MOG2 adapts to background change better than Frame Differencing in some conditions.
- **H4:** Stabilizing before background subtraction can improve foreground quality on camera-jitter video.
- **H5:** Removing some DSP stages changes pipeline quality and performance.
- **H6:** The order of image transformations influences segmentation results.
- **H7:** The configuration with the highest quality is not necessarily the one with the best computational performance.

As required by the project design, these hypotheses are tested empirically and are not assumed true.

7. Literature Review

Camera-motion estimation for stabilization builds on sparse feature tracking. Lucas and Kanade (1981) introduced an iterative registration technique that uses spatial intensity gradients and Newton–Raphson iteration to estimate local displacement, and Shi and Tomasi (1994) proposed selecting corners by the minimum eigenvalue of the 2×2 structure tensor so that the selected features are exactly those the tracker can follow. Because independently moving objects produce gross outliers among the tracked correspondences, robust estimation is required; Fischler and Bolles (1981) introduced random sample consensus (RANSAC), which fits a model to minimal subsets and keeps the hypothesis with the largest consensus set.

Dense alignment is an alternative to sparse tracking. Evangelidis and Psarakis (2008) maximize the enhanced correlation coefficient, a zero-mean normalized cross-correlation that is invariant to global brightness and contrast change, over a parametric warp. Once frame-to-frame motion is known, stabilization amounts to smoothing the accumulated camera path. Grundmann et al. (2011) compute optimal L1 camera paths that separate intentional motion from shake under crop constraints, and Liu et al. (2013) extend this to spatially variant "bundled" camera paths on a mesh so that depth parallax and rolling shutter can be handled, which a single global transform cannot do. Zhang et al. (2019) argue that stabilization must be assessed with objective motion-stability measures rather than visual inspection alone.

For background modeling, Piccardi (2004) reviewed the main families, from frame differencing and running averages to mixture models and non-parametric density estimators. Stauffer and Grimson (1999) modeled each pixel with an adaptive mixture of Gaussians, Zivkovic (2004) made the number of components adaptive (the algorithm known as MOG2 in OpenCV), and Zivkovic and van der Heijden (2006) proposed a non-parametric K-nearest-neighbor density estimate per pixel. Barnich and Van Droogenbroeck (2011) introduced ViBe, which stores a small set of past samples per pixel and refreshes them by random replacement. Chapel and Bouwmans (2021) survey the harder case of moving-object detection with a moving camera, in which global motion compensation precedes or accompanies background modeling.

Benchmarking background subtraction requires pixel-accurate ground truth; the CDnet 2014 dataset (Wang et al., 2014) provides 22 additional videos with about 70,000 annotated frames across five challenge categories, including camera jitter. Appendix A summarizes these works in a literature matrix.

8. Research Gap

The reviewed works typically propose or analyze one algorithm family at a time. Stabilization papers evaluate trajectory smoothness or visual quality, and background-subtraction papers assume a static or already compensated camera. Within the studies reviewed here, there is little controlled evidence that isolates how each classical DSP stage (feature selection, outlier rejection, trajectory smoothing, warping, shadow handling, morphology) contributes when the stages are chained, and little that distinguishes ground-truth metrics from proxy metrics in a single reproducible tool. This project addresses that gap with a unified workbench in which the same frames pass through all algorithms with identical settings.

9. Expected Contributions

This project provides the following contributions:

- **A transparent browser-native workbench:** OpenCV.js WebAssembly plus verified TypeScript routines (2D similarity RANSAC with Procrustes refinement, a Euclidean ECC solver, trajectory smoothing, a project Sample-based KNN, morphology and connected components), with explicit runtime capability inspection and fallbacks.
- **A metric framework that refuses false precision:** IoU, precision, recall, F1 and Transform RMSE are reported only with ground truth; otherwise the application prints "N/A – Ground Truth Unavailable".
- **A research module:** five experiment groups (A stabilization, B background, C processing order, D ablation, E parameter sensitivity), statistical utilities and an academic report builder.
- **A licensed 15-video dataset with documented measurements:** three categories of five videos each, with sources, authors, licenses and measured jitter/foreground statistics.

10. Methodology

10.1. Research Framework Workflow

[Video input: MP4 / MOV / WebM, or Synthetic Demo Scenario]
↓
[1. Decoding: sequential frames, ≤ 90 frames, ≤ 480 px width, 15 fps, ≤ 12 s]
↓
[2. Module A: Stabilization (A0 Original | A1 LK + RANSAC | A2 ECC)
→ camera trajectory → Gaussian smoothing → corrective warp]
↓
[3. Module B: Background / Foreground (B1 Frame Diff. | B2 MOG2 | B3 Sample KNN)
→ morphology (blur, threshold, opening, closing, area filter)]
↓
[4. Module C: Combined pipeline (Flow A: Original → B vs Flow B: Stabilized → B)]
↓
[5. Module D: Evaluation (ground-truth, proxy and runtime metrics)]
↓
[6. Research Lab: ablation, sensitivity, statistics, report export]

10.2. Software Architecture

The application is a React 19 + TypeScript + Vite single-page app styled with Tailwind CSS. Computer-vision routines run on `@techstark/opencv-js` 5.0.0-release.1 (WebAssembly) in a Web Worker, with Recharts for visualization and JSZip for importing ground-truth mask archives. The source tree separates concerns as follows (Table 10.1).

Table 10.1. Source-code organization of the SpatioTemporal Video Processing Lab.

Directory / file	Responsibility
algorithms/stabilization/	opticalFlow.ts (Shi–Tomasi + pyramidal LK), ransac.ts (2D similarity RANSAC + Procrustes), ecc.ts (Euclidean ECC), trajectory.ts (accumulation + Gaussian smoothing), stabilizationPipeline.ts (warp, border loss, orchestration)
algorithms/background/	frameDifference.ts, mog2.ts, sampleKnn.ts, morphology.ts, foregroundExtraction.ts (five display modes)
core/	openCvLoader.ts (WASM init and capability inspection), videoDecoder.ts, videoEncoder.ts (MediaRecorder), workerBridge.ts, pipelineManager.ts (8-stage combined pipeline)
evaluation/	stabilizationMetrics.ts, segmentationMetrics.ts, performanceMetrics.ts, benchmarkRunner.ts, researchExperiments.ts (groups A–E), statisticalAnalysis.ts, reportGenerator.ts
demo/	syntheticSceneGenerator.ts (3 scenarios), groundTruthGenerator.ts (exact masks, GT ZIP loader)
pages/ and research/	Dashboard, Stabilization Lab, Background Lab, Combined Pipeline, Benchmark Lab, Research Lab; literature database and report builder

10.3. Video Input and Resource Guardrails

Frames are decoded sequentially from an HTML5 video element. The defaults in `DEFAULT_VIDEO_INPUT_CONFIG` are 90 frames, 480 px maximum width, 15 fps target sampling and 12 s maximum duration, which bound memory use and keep WebAssembly processing responsive. All algorithms in one comparison receive exactly the same decoded frame array (identical resolution, frame count and sampling rate).

10.4. Module A: Video Stabilization

10.4.1. A1: Pyramidal Lucas–Kanade Optical Flow + 2D Similarity RANSAC

Each frame is converted to 8-bit luminance (ITU-R BT.601). Up to `maxCorners` = 180 Shi–Tomasi corners are detected (`qualityLevel` = 0.015, `minDistance` = 8 px) and tracked from frame t−1 to t with pyramidal Lucas–Kanade (15 × 15 window, two pyramid levels beyond the base). The inter-frame camera motion is the four-degree-of-freedom similarity transform

$$x' = s \cdot R(\theta) \cdot x + (tx, ty)^T$$

estimated by RANSAC with two-point minimal subsets and an inlier threshold of 2.5 px, followed by a closed-form Procrustes least-squares refinement over all inliers (`ransac.ts`). The translation is converted to the displacement of the image center, so that rotation about the center is not confused with translation, and the per-frame increments (dx, dy, dθ) are accumulated into the camera trajectory P(t).

10.4.2. A2: Enhanced Correlation Coefficient (ECC) Euclidean Alignment

ECC maximizes the zero-mean normalized correlation between consecutive frames over a three-degree-of-freedom Euclidean motion (tx, ty, θ). The project's TypeScript solver (`ecc.ts`) works on a 2× subsampled grid with a 10-pixel margin, uses at most 25 Gauss–Newton iterations and stops at `eccEps` = 0.001. When it does not converge, the frame is recorded in `eccFailedFrames`, the identity transform is used and the ECC convergence rate decreases accordingly.

10.4.3. Trajectory Smoothing and Corrective Warp

The cumulative trajectory $P(t) = \Sigma k\ \Delta T(k)$ is smoothed per component by a normalized one-dimensional Gaussian kernel of radius R (default R = 10, $\sigma = \max(0.8, R/2.2)$, reflective boundary). The corrective transform is the difference between the smoothed and raw trajectory:

$$\Delta C(t) = P_smooth(t) - P_raw(t)$$

Each frame is warped around the image center by ΔC(t) with bilinear interpolation; out-of-range source pixels replicate the nearest border pixel so that moving black borders do not create false foreground in the subsequent background model. The valid-area ratio of every warp is recorded to report border loss.

10.5. Module B: Background / Foreground Separation

Three motion-based models are compared (this is motion segmentation, not semantic recognition of people or vehicles).

- **B1 Frame Differencing:** $M(x, y, t) = 255$ if $|I(x, y, t) - B(x, y)| \geq T$, else 0, with T = 28. The default reference B is the temporal median of the first five frames; first-frame and exponential running-average references are also available.
- **B2 MOG2:** per-pixel adaptive Gaussian mixture (Zivkovic, 2004) with history 60, variance threshold 16, learning rate 0.03 and shadow detection enabled (shadow pixels are labeled 127 and excluded from the binary foreground).
- **B3 Project Sample-based KNN:** each pixel stores N = 14 grayscale samples; a pixel is background if at least K = 3 samples lie within distance R = 24, and samples are refreshed stochastically with learning rate 0.08. It is a project implementation, not OpenCV's `BackgroundSubtractorKNN`, which standard OpenCV.js builds omit.

All masks then pass through the same post-processing (`morphology.ts`): 3 × 3 Gaussian blur, re-thresholding at 127, 3 × 3 opening, 3 × 3 closing and 8-connected-component filtering that removes blobs smaller than 28 px² and extracts bounding boxes.

10.6. Module C: Combined Spatiotemporal Pipeline

The combined pipeline compares **Flow A** (original frames → background subtraction) with **Flow B** (stabilized frames → background subtraction) on the same input. Camera shake in Flow A makes

background edges appear as foreground; Flow B removes most of this motion first. The pipeline manager (`pipelineManager.ts`) executes eight stages and records metrics for each flow.

10.7. Evaluation Metrics

Metrics are grouped by evidential strength so that proxies are never presented as accuracy.

- **Stabilization (proxy, real video):** translation Jitter RMS and rotation Jitter RMS, defined as the RMS of the second-order finite difference of the trajectory,

$$\Delta^2 P(t) = P(t+1) - 2P(t) + P(t-1); \quad \text{Jitter RMS} = \sqrt{\left(\frac{1}{N-2}\right) \Sigma \left(\Delta^2 P(t)\right)^2}$$

- **Stabilization (other):** RANSAC inlier ratio, ECC convergence rate, border loss = 100 % − valid-area %, processing FPS.
- **Segmentation (ground truth required):** TP, FP, FN, TN, IoU = TP/(TP+FP+FN), precision, recall, F1, with zero-division safeguards. Frame 0 is excluded from sequence averages.
- **Segmentation (proxy, always available):** foreground ratio and temporal mask flicker (percentage of pixels that change state between consecutive masks). Foreground ratio is never interpreted as accuracy.
- **Performance:** median of three trials, ms per frame and FPS, with warm-up and export time excluded from algorithm time.

11. Experimental Design

11.1. Dataset

The dataset contains **15 short videos (4–10 s), five per category**, kept at or below 640 px width and encoded as H.264 MP4 without audio. All videos come from Pexels (Pexels License: free use without mandatory attribution; authors are nevertheless credited in Appendix D) and are shared in a Google Drive folder (https://drive.google.com/drive/folders/1jula3nmZzVjUM8j76i8jq4CKwTAkWs7k). They were chosen by the following criteria:

- **Stabilization (01_Stabilization):** visible camera shake over a mostly static scene, covering mild to severe shake (measured jitter 3.3 to 14.2 px) so that algorithm behavior can be compared across severity.
- **Background subtraction (02_Background_Subtraction):** fixed camera (measured jitter ≤ 0.4 px) with moving vehicles and pedestrians; one clip (a time-lapse of moving clouds) adds a dynamic background and illumination drift.
- **Combined (03_Combined):** simultaneous camera motion and moving foreground, from slight handheld shake to severe shake.

Each file was measured with an independent OpenCV script (LK + RANSAC trajectory, high-pass residual after Gaussian smoothing, on 320-px-wide frames) to confirm that it belongs to its category. Table 11.1 lists the final set.

Table 11.1. The 15-video dataset (jitter and moving-pixel ratio are measured proxy values, not ground truth).

Video	Category	Res.	Dur. (s)	Jitter (px)	Moving px (%)	Content
stab_01	Stabilization	640x360	10.0	3.32	4.3	POV walking on fall leaves
stab_02	Stabilization	640x360	10.0	3.50	5.2	POV walking on wet pavement, splashes
stab_03	Stabilization	640x360	10.0	6.06	14.7	POV walking on sunlit brick pavement with shadow
stab_04	Stabilization	640x360	10.0	10.41	14.3	POV walking on concrete sidewalk, then grass
stab_05	Stabilization	640x360	5.7	14.17	27.0	First-person walking on a dirt path along grass
bg_01_i	Background	640x360	10.0	0.11	4.0	Montreal street intersection, cars and pedestrians
bg_02_w	Background	640x360	10.0	0.11	1.9	Montreal winter street, pedestrians
bg_03_l	Background	640x360	10.0	0.18	3.1	London street, pedestrians, buses, taxis
bg_04_t	Background	640x360	10.0	0.38	6.3	Snowy street, green electric bus, cars (high angle)
bg_05_t	Background	640x360	10.0	0.37	1.3	Time-lapse of cars and moving clouds at an intersection
comb_01	Combined	640x360	4.0	21.85	29.2	Shaky handheld footage of people walking in a winter city
comb_02	Combined	640x360	5.0	1.56	12.9	High aerial shot of vehicles at an intersection
comb_03	Combined	640x360	6.2	1.61	8.5	Night street celebration, crowd holding phones and flags
comb_04	Combined	640x360	9.4	1.21	11.8	Crowd crossing a downtown commercial district (low camera)
comb_05	Combined	360x640	10.0	12.65	11.3	Runner on a street in a small European town (portrait)

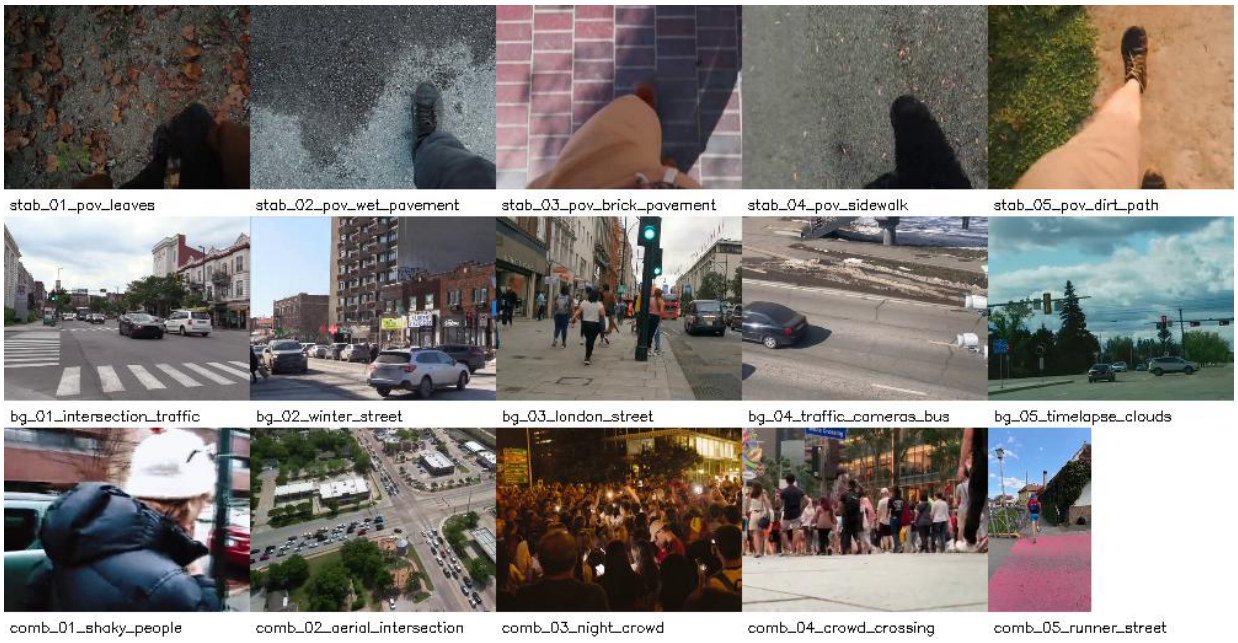


Figure 11.1. Sample frame from each of the 15 dataset videos (rows: stabilization, background subtraction, combined).

No ground-truth foreground masks exist for these real videos. In keeping with the application's rule, IoU, precision, recall and F1 are therefore **not computed** for them ("N/A – Ground Truth Unavailable"), and segmentation is compared only through proxy metrics (Section 12.4). Ground truth is used where it exists: in the controlled validation of Section 11.2.

11.2. Controlled Validation with Known Camera Motion

To validate the stabilization estimators against exact ground truth, two static-camera videos (bg_01 and bg_02) were shaken synthetically: frame t is warped by a known transform consisting of harmonic horizontal and vertical translation (1.8 Hz and 1.3 Hz), harmonic roll (1.1 Hz) and Gaussian noise, mirroring the project's Scenario 1 generator. Three severities were used (translation amplitude / roll amplitude: Low 1.5 px / 0.25°, Medium 3.6 px / 0.6°, High 6.0 px / 1.0°) plus a static control with no added motion. Because the true transforms are known, the inter-frame Transform RMSE can be computed in addition to the jitter.

11.3. Independent & Dependent Variables

- **Independent variables:** stabilizer (A0, A1, A2); background model (B1, B2, B3); processing flow (A or B); ablated stage (D0–D6); smoothing radius R ∈ {3, 6, 10, 15, 20}; MOG2 variance threshold ∈ {8, 16, 25, 36, 50}; shake severity (controlled data).
- **Dependent variables:** translation/rotation Jitter RMS (px, deg), Transform RMSE (px, deg; controlled data only), RANSAC inlier ratio, border loss (%), foreground ratio (%), mask flicker (%), connected-component count, ms per frame and FPS.

11.4. Ablation Pipeline Matrix

The ablation follows the project's Group D definition: each variant removes one stage of the full pipeline while all other parameters and the input frames are identical.

Table 11.2. Ablation variants (Group D).

Variant	Stage bypassed	How it is bypassed
D0 Full pipeline	—	LK + RANSAC → Gaussian smoothing (R = 10) → warp → MOG2 with shadow detection → blur + opening + closing + area filter
D1	Spatial denoising	3 × 3 Gaussian blur before re-thresholding disabled
D2	RANSAC outlier rejection	All tracked correspondences accepted (threshold 9999 px); least-squares fit on all points
D3	Trajectory smoothing	Smoothing radius 0, so the smoothed trajectory equals the raw one and the correction is zero
D4	Motion compensation	Warping bypassed; unstabilized frames go directly to MOG2
D5	Morphological processing	Opening, closing and area filter disabled (raw MOG2 mask)
D6	Shadow suppression	MOG2 `detectShadows` = false (shadows classified as foreground)

Note that in the application's definition D3 and D4 coincide numerically: with radius 0 the corrective transform is the zero transform, so the frames passed to MOG2 are unchanged. The two rows are kept for fidelity to the project design and should match in the results.

11.5. Experimental Protocol and Reference Implementation

All frame sequences were decoded exactly as the application does (first 90 frames, 15 fps, at most 480 px wide, at most 12 s). All algorithms receive the same frames. Frame 0 is excluded from foreground-ratio and flicker averages.

Important disclosure. The numerical results in Section 12 were produced by a **Python/OpenCV reference re-implementation** of the application's algorithms that uses the application's default parameters (`src/utils/defaultParams.ts`), the same trajectory-smoothing kernel, the same corrective-warp convention, the same Jitter RMS and border-loss definitions and the same morphology settings. It was written first, because the browser application could not be run at that stage, and it is still used for the synthetic-motion, ablation and sensitivity experiments, which need controls the application does not expose. The browser application was afterwards installed and run on all 15 videos (Section 12.8). The reference implementation is not the browser application itself: LK/RANSAC use OpenCV's `calcOpticalFlowPyrLK` and `estimateAffinePartial2D` (the application uses OpenCV.js plus its own TypeScript RANSAC), ECC uses OpenCV's `findTransformECC` on a half-resolution grid with 25 iterations (the application uses its own TypeScript solver), B3 is a vectorized port of `sampleKnn.ts`, and

timings are for native Python/OpenCV, not WebAssembly. Absolute timings and some ECC/KNN details therefore differ from the application's Benchmark Lab export, and Section 12.8 reports the two side by side where they can be compared. Timings are the median of three trials from a separate, otherwise idle run; all other metrics are deterministic and were computed once.

12. Experimental Results

All values in Sections 12.1–12.7 are read from the JSON output of the reference implementation (Section 11.5); Section 12.8 reports the browser application's own export. "Jitter" is the translation Jitter RMS (second-order finite-difference RMS of the cumulative camera trajectory, in pixels at the processing resolution). Where it is labeled "re-measured", the jitter of the output video was estimated again with the same LK + RANSAC estimator applied to the stabilized frames, which is stricter than reading the smoothed trajectory because it detects residual motion that the stabilizer failed to remove.

12.1. Dataset Characterization

Figure 12.1 plots every video by its measured jitter and moving-pixel ratio. The three categories separate as designed: the five background-subtraction videos lie at jitter ≤ 0.4 px, the stabilization videos span about 3 to 14 px, and the combined videos combine appreciable motion statistics with moving foreground, from slight handheld shake (about 1 px) to severe shake (more than 20 px).

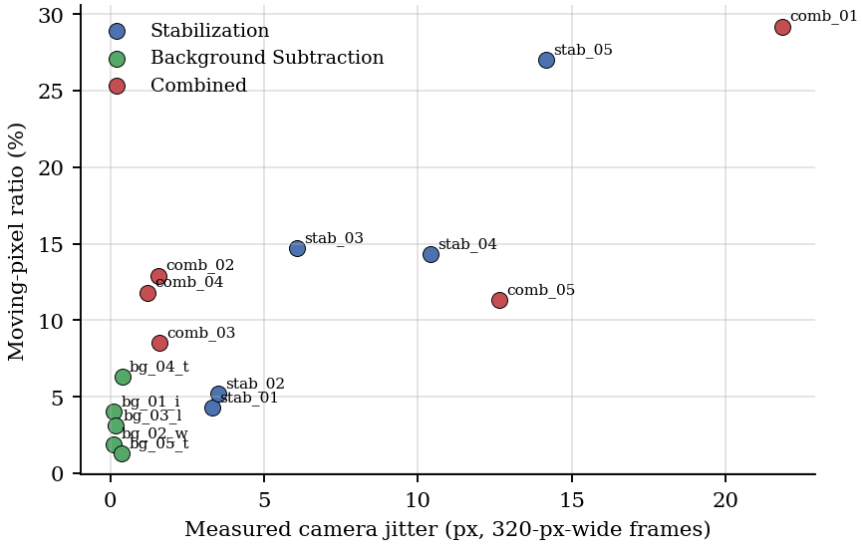


Figure 12.1. Measured camera jitter versus moving-pixel ratio for the 15 dataset videos.

Two honest qualifications apply. First, the stabilization videos are point-of-view walking shots in which the ground texture dominates the frame and parts of the body are visible; this is shake with strong parallax and a partly camera-attached object, which is a harder case than a pure handheld pan. Second, three of the five "combined" videos (comb_02 to comb_04) have only slight camera motion (jitter about 1.2 to 1.6 px), so they test the combined pipeline in a mild regime; only comb_01 and comb_05 are severe.

12.2. Controlled Validation with Known Camera Motion

Table 12.1 reports the controlled experiment of Section 11.2, averaged over the two source videos. Jitter is re-measured on the shaken input ("before") and on the stabilized output ("after").

Table 12.1. Stabilization of synthetically shaken video with known camera transforms (mean of two videos).

Shake level	Method	True jitter (px)	Before (px)	After (px)	Reduction	Transform RMSE (px)	Rot. RMSE (deg)	ms / frame
Low	A1 LK+RANSAC	0.92	0.93	0.12	87 %	0.14	0.015	6.8
	A2 ECC	0.92	0.93	0.13	86 %	0.23	0.040	7.4
Medium	A1 LK+RANSAC	2.21	2.21	0.11	95 %	0.14	0.016	7.5
	A2 ECC	2.21	2.21	0.14	94 %	0.23	0.040	8.8
High	A1 LK+RANSAC	3.68	3.68	0.14	96 %	0.14	0.015	9.0
	A2 ECC	3.68	3.68	0.14	96 %	0.24	0.041	9.2
Static control	A1 LK+RANSAC	0.00	0.08	0.10	—	0.13	0.016	7.7
	A2 ECC	0.00	0.08	0.13	—	0.23	0.040	6.4

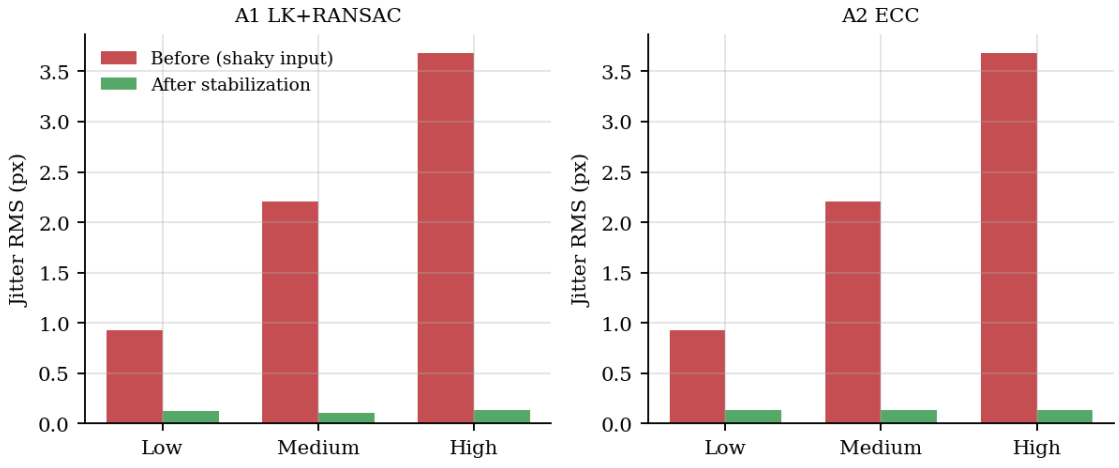


Figure 12.2. Residual jitter before and after stabilization on synthetically shaken video.

With ground truth, both estimators remove most of the injected shake: at the High level the jitter falls from 3.68 px to 0.14 px (A1) and 0.14 px (A2). The inter-frame transform error is small for both (mean translation RMSE 0.14 px for A1 and 0.24 px for A2), and A1 is slightly more accurate than A2 on these

clips. On the static control neither method adds appreciable motion (residual jitter stays near the level of the original video). This supports **H1** (smoothing reduces trajectory oscillation) in the regime where tracking is reliable. These shake patterns are smooth and low-amplitude, so the result does not transfer automatically to severe real shake, as Section 12.3 shows.

12.3. Stabilization on Real Videos (Module A)

Table 12.2. Re-measured translation jitter RMS (px) and tracking statistics on the ten shaky videos.

Video	A0 Original	A1 LK+RANSAC	A2 ECC	A1 RANSAC inliers	A1 border loss (%)	A2 border loss (%)
stab_01_pov_leaves	3.58	0.43	0.77	94 %	1.3	1.2
stab_02_pov_wet_pavement	3.80	0.41	1.96	95 %	1.5	1.4
stab_03_pov_brick_pavement	6.75	0.72	12.22	60 %	2.3	2.6
stab_04_pov_sidewalk	11.56	1.59	12.53	45 %	4.2	2.9
stab_05_pov_dirt_path	23.24	13.38	24.74	30 %	5.9	5.7
comb_01_shaky_people	9.03	6.77	7.30	42 %	7.0	7.3
comb_02_aerial_intersection	0.79	0.51	0.35	89 %	0.3	0.2
comb_03_night_crowd	0.88	0.21	0.47	85 %	0.3	0.4
comb_04_crowd_crossing	0.88	0.50	0.62	84 %	0.3	0.3
comb_05_runner_street	9.06	8.74	8.32	50 %	2.2	1.0

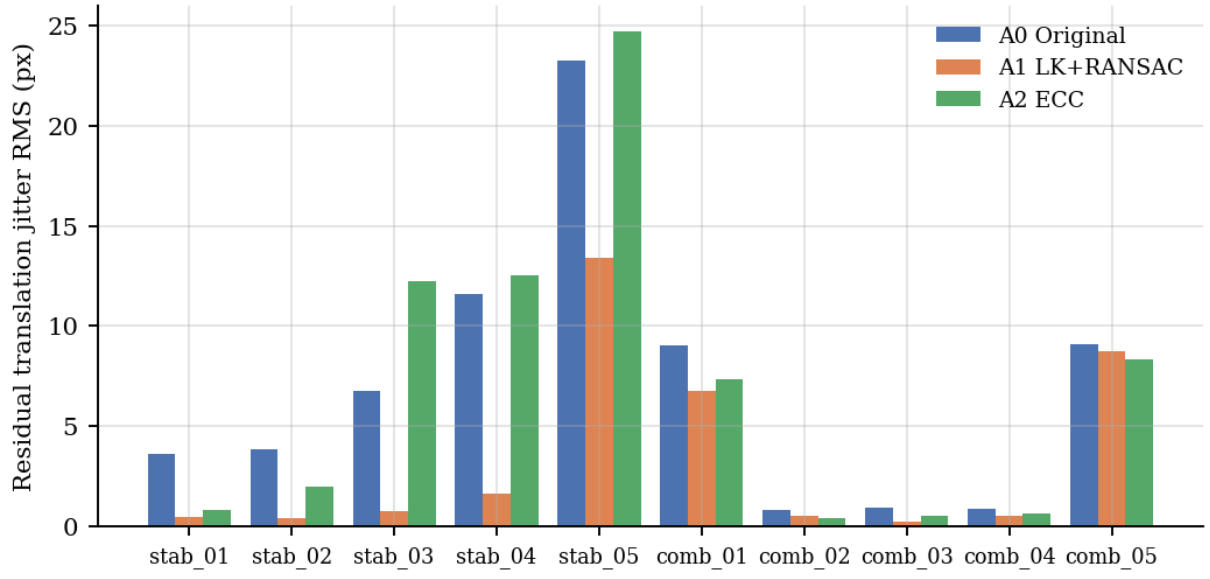


Figure 12.3. Re-measured residual translation jitter of A0, A1 and A2 on the ten shaky videos.

A1 (LK + RANSAC) reduces the re-measured jitter by 88 % (stab_01), 89 % (stab_02), 89 % (stab_03) on the three mildest stabilization clips and by 86 % on stab_04. On the most severe clips the benefit shrinks sharply: 42 % on stab_05 (inlier ratio 30 %) and only 25 % on comb_01 (inlier ratio 42 %) and 4 % on comb_05. Across the ten shaky videos the Pearson correlation between A1's RANSAC inlier ratio and its jitter reduction is +0.39 (n = 10), a modest association: low inlier ratios accompany the weak results on stab_05, comb_01 and comb_05, where large moving foreground or strong parallax leaves few points that agree on one global motion, but stab_04 improved by 86 % despite an inlier ratio of 45 %. The inlier ratio is therefore a warning sign, not a rule.

A2 (ECC) behaves differently. It lowers jitter on the mildest clips (for example stab_01: 3.58 → 0.77 px) but **increases** the re-measured jitter on stab_03, stab_04 and stab_05 (6.75 → 12.22, 11.56 → 12.53 and 23.24 → 24.74 px). A single-scale ECC with a limited iteration budget has a small capture range: for displacements of several pixels per frame, or for the periodic brick texture of stab_03, it settles in a wrong local optimum and the smoothing then amplifies the estimation error. Its trajectory-based jitter (the smoothed-trajectory value that the application would display) nevertheless looks excellent, which illustrates why a proxy computed from the algorithm's own trajectory cannot by itself prove that stabilization worked. This supports **H2**: the two methods behave differently depending on the motion type and the content.

A caveat on fairness: the re-measurement uses LK + RANSAC, the same estimator family as A1, so it may favor A1. The controlled experiment of Section 12.2 provides the unbiased comparison against ground truth, and there the two methods are close. The disagreement on real severe shake is therefore best read as ECC failing on large frame-to-frame displacement rather than as a general superiority of A1.

Cost: the median of three timed trials (algorithm time only, averaged over the 15 videos) was 6.6 ms/frame for A1 (151.3 FPS) and 9.1 ms/frame for A2 (110.0 FPS) in native Python/OpenCV at up to 480 px width. These absolute values are not comparable with WebAssembly timings; only the ordering is meaningful. Border loss grows with shake severity (A1: 1.3 % on stab_01 up to 5.9 % on stab_05 and 7.0 % on comb_01) and is close to zero for the static-camera clips.

12.4. Background / Foreground Separation (Module B)

Table 12.3 compares the three background models on the five fixed-camera videos. Without ground truth, only proxy metrics are available: the foreground ratio and the temporal flicker of the cleaned masks. **These cannot rank accuracy**: a low foreground ratio may mean a clean mask or a missed object. They are informative mainly when the scene is known to contain little motion.

Table 12.3. Foreground ratio (FG, %) and mask flicker (FL, %) of B1, B2 and B3 on the fixed-camera videos (cleaned masks).

Video	B1 FG	B1 FL	B2 FG	B2 FL	B3 FG	B3 FL
bg_01_intersection_traffic	12.7	2.7	1.9	2.1	12.6	2.5
bg_02_winter_street	10.5	1.3	1.1	1.1	10.8	1.3
bg_03_london_street	13.1	2.9	2.2	2.3	13.7	3.0

Video	B1 FG	B1 FL	B2 FG	B2 FL	B3 FG	B3 FL
bg_04_traffic_cameras_bus	28.9	5.0	3.4	3.2	32.4	5.0
bg_05_timelapse_clouds	19.4	2.3	0.4	0.6	20.2	2.2
Median ms / frame (3 trials)	0.07		2.87		15.21	

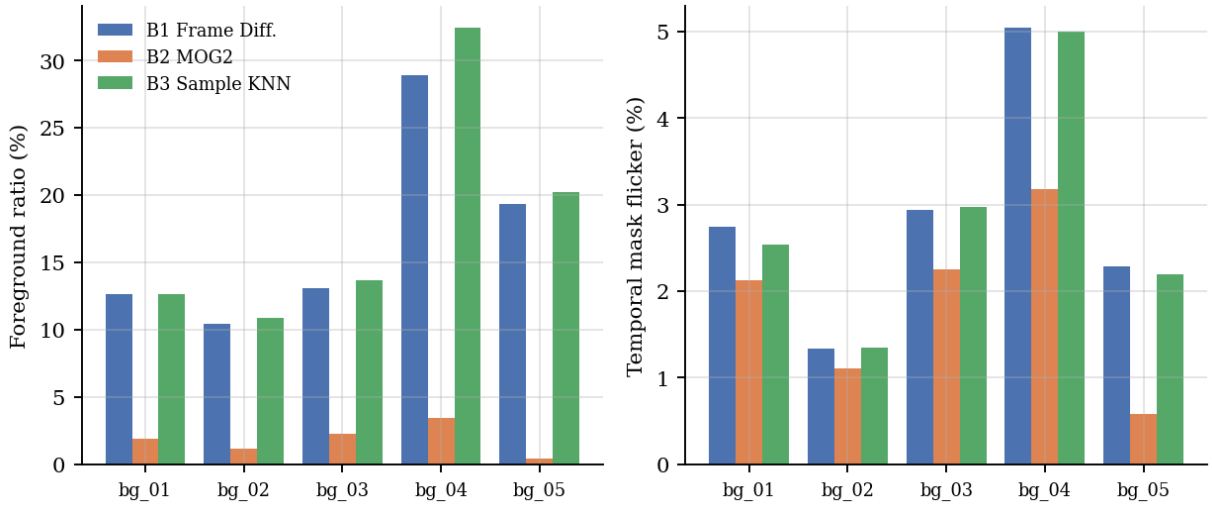


Figure 12.4. Foreground ratio and mask flicker of B1, B2 and B3 on the five fixed-camera videos.

The clearest separation is on the time-lapse of moving clouds (bg_05), a scene with essentially no foreground object but a continuously changing sky: B1 and B3 label 19.4 % and 20.2 % of the frame as foreground whereas B2 labels 0.4 %. Over all five fixed-camera videos the mean foreground ratio is 16.9 % (B1), 1.8 % (B2) and 18.0 % (B3). B1 uses a fixed reference taken from the first five frames and therefore cannot follow illumination drift, which is consistent with the literature on frame differencing (Piccardi, 2004) and with **H3**. B3 behaves almost like B1 here because its sample bank is initialized from the first frame with ±1 variation and refreshed only slowly (learning rate 0.08), so with the default K = 3 and R = 24 it also lags behind sustained change. That the proxy supports H3 is encouraging, but H3 is strictly **not confirmed**: without masks we do not know whether B2's small foreground ratio is accuracy or under-segmentation of slow or stopped objects.

On speed the ordering is unambiguous: median 0.07 ms/frame (B1), 2.87 ms (B2) and 15.21 ms (B3). B1 is the cheapest and B3, which scans the whole sample bank per pixel, is the most expensive.

12.5. Combined Pipeline: Flow A versus Flow B (Module C)

Flow A feeds the original frames to MOG2; Flow B feeds LK + RANSAC-stabilized frames to MOG2; Flow B2 uses ECC-stabilized frames. Table 12.4 reports the proxy metrics on the five combined videos.

Table 12.4. Foreground ratio (FG, %) and mask flicker (FL, %) of the three processing flows on the combined videos.

Video	A FG	A FL	B FG	B FL	B2 FG	B2 FL	ΔFL (B−A)
comb_01_shaky_people	27.8	27.3	26.6	25.9	26.2	25.3	-1.3
comb_02_aerial_intersection	5.4	5.2	5.0	4.8	4.7	4.5	-0.4
comb_03_night_crowd	6.0	6.0	5.4	5.4	5.4	5.3	-0.6
comb_04_crowd_crossing	8.2	7.4	7.8	7.1	7.9	7.1	-0.4
comb_05_runner_street	10.8	10.8	10.8	10.9	10.0	10.2	+0.2

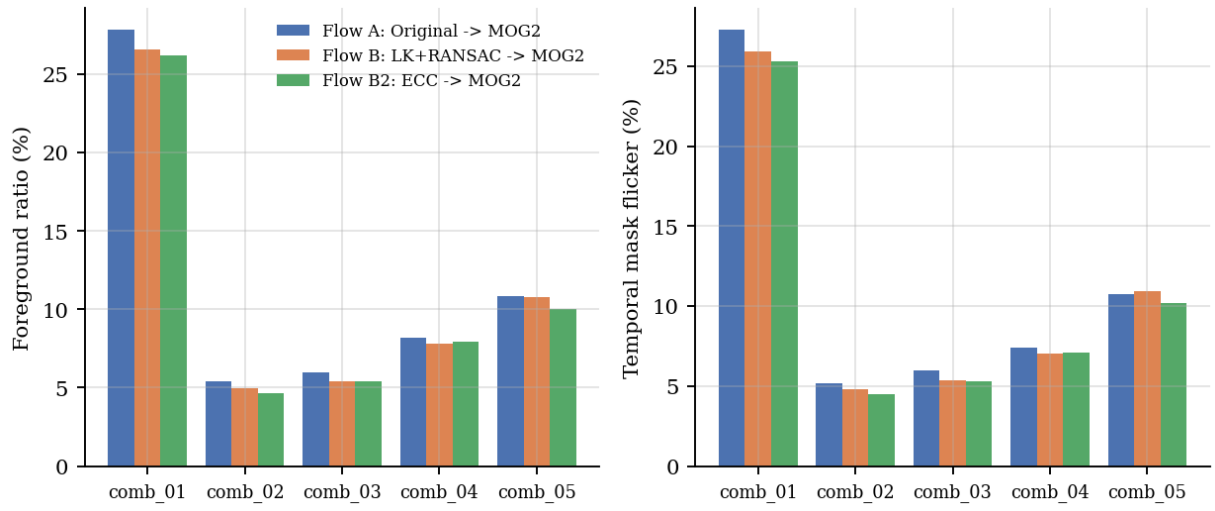


Figure 12.5. Foreground ratio and mask flicker for Flow A, Flow B and Flow B2 on the five combined videos.

Stabilizing first changes the proxy metrics only slightly: the mean flicker difference (Flow B minus Flow A) is -0.50 percentage points and the mean foreground-ratio difference is -0.53 points. Flicker decreases on 4 of the five combined videos; the largest reduction is on comb_01_shaky_people (-1.3 points) and the only increase is on comb_05_runner_street. On comb_01, where stabilization itself only partly worked (RANSAC inliers 42 %), flicker is still 25.9 % after stabilization.

This is **weak, proxy-only support for H4** and not a demonstration of it. The differences are of the order of one percentage point, they are measured without ground truth, and a lower flicker or foreground ratio is not by itself evidence of a more accurate mask. On the static-camera and stabilization videos the two flows are practically identical (Appendix tables of the JSON output), which is the expected result when stabilization has nothing to correct and also indicates that the stabilizer does not damage a stable video. A decisive test of H4 and H6 requires ground-truth masks, for example the CDnet 2014 camera-jitter category, or the application's Scenario 1, which provides exact masks.

12.6. Ablation Study (Group D)

Table 12.5 averages the seven ablation variants over the five combined videos. Every variant uses the same frames and parameters except the bypassed stage (Table 11.2).

Table 12.5. Ablation results, mean over the five combined videos (foreground ratio and flicker in %, jitter in px).

Variant	FG ratio	Flicker	Components / frame	Re-measured jitter	Border loss
D0 Full	**11.1**	**10.8**	**58**	**3.35**	**2.0**
D1 No denoise	7.5	8.4	46	3.35	2.0
D2 No RANSAC	11.6	11.2	59	6.34	2.4
D3 No smoothing	11.6	11.3	58	4.13	0.0
D4 No motion comp.	11.6	11.3	58	4.13	—
D5 No morphology	16.3	15.8	1745	3.35	2.0
D6 No shadow supp.	22.4	15.9	66	3.35	2.0

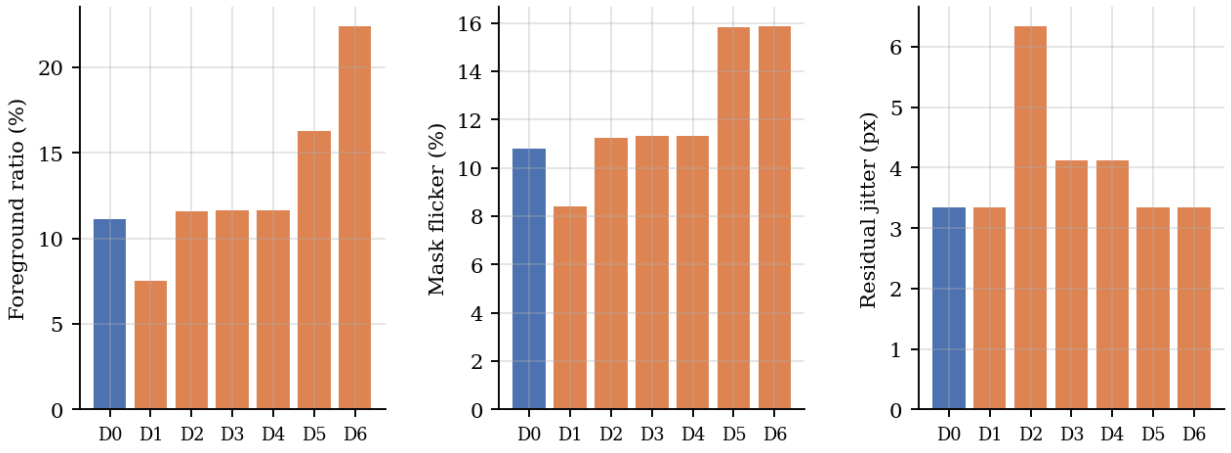


Figure 12.6. Ablation impact on foreground ratio, mask flicker and residual jitter (mean of five combined videos).

The stages differ greatly in impact. **Removing shadow suppression (D6)** more than doubles the foreground ratio (11.1 % → 22.4 %) and raises flicker to 15.9 %, because cast shadows are labeled foreground. **Removing morphological processing (D5)** raises flicker to 15.8 % and multiplies the number of connected components per frame from 58 to 1745, i.e. the raw MOG2 mask is dominated by isolated noise pixels. **Removing RANSAC (D2)** nearly doubles the re-measured jitter (3.35 → 6.34 px) and the damage is concentrated on the video with the most independent motion (comb_01: 6.77 → 17.38 px), where moving people bias the least-squares camera estimate. **Removing smoothing (D3) or motion**

compensation (D4) gives identical results (11.3 % flicker for both), as predicted in Section 11.4, and a mask flicker only 0.5 percentage points above the full pipeline: on these mostly mild-shake videos, geometric stabilization has a small effect on the proxy mask metrics compared with shadow handling and morphology.

One result runs against the expectation. **Removing the 3 × 3 spatial blur (D1) lowered** both the foreground ratio (7.5 %) and the flicker (8.4 %) relative to the full pipeline. A blur followed by re-thresholding at 127 slightly widens and merges foreground blobs in this implementation, so on proxy metrics the extra denoising does not help; whether it improves true IoU cannot be decided without ground truth. This supports **H5** (removing stages changes quality) but shows that "more DSP" is not automatically better.

12.7. Parameter Sensitivity (Group E)

Two parameters were swept one at a time (five values each, all others at default): the trajectory smoothing radius R on two shaky videos, and the MOG2 variance threshold on two fixed-camera videos.

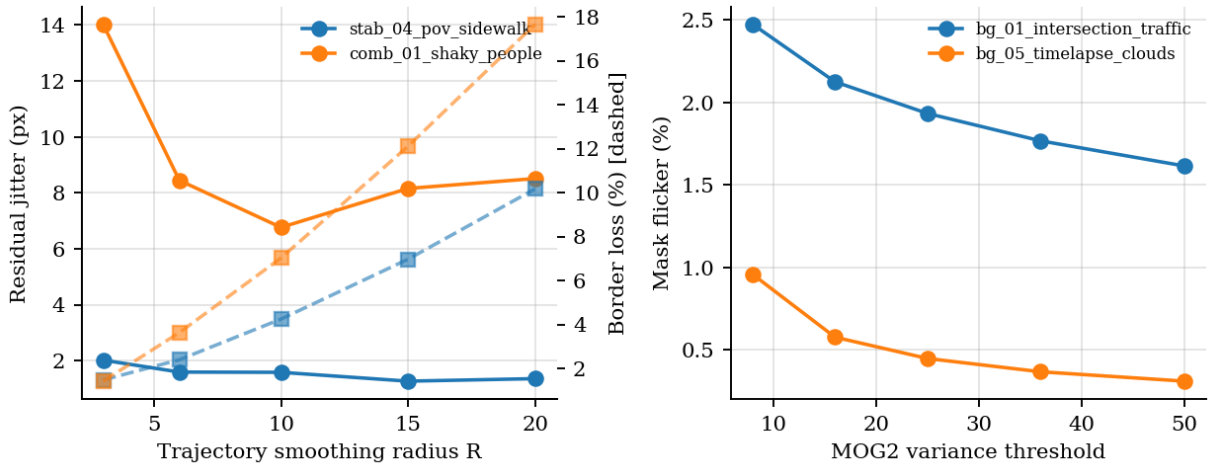


Figure 12.7. Left: smoothing radius versus re-measured jitter (solid) and border loss (dashed). Right: MOG2 variance threshold versus mask flicker.

Smoothing radius. The jitter of the smoothed trajectory falls monotonically with R (stab_04: 1.90 → 0.76 px; comb_01: 4.30 → 1.45 px), but the **re-measured** jitter of the stabilized video does not: it is lowest at R = 15 for stab_04 (1.27 px) and at R = 10 for comb_01 (6.77 px) and worsens afterwards. Border loss grows steeply with R (1.5 % → 10.2 % on stab_04, 1.4 % → 17.7 % on comb_01). A radius of about 10 to 15, close to the application default of 10, is therefore a reasonable compromise between residual shake and lost image area.

MOG2 variance threshold. Raising the threshold monotonically reduces the foreground ratio, flicker and component count (bg_01 flicker 2.47 % → 1.61 %, bg_05 0.96 % → 0.31 %), because the model becomes less sensitive; processing time is unaffected. As always for proxy metrics, the best value cannot be chosen from these curves without ground truth, since a very high threshold would also suppress genuine foreground.

12.8. Results from the Browser Application (Benchmark Lab Export)

After the reference-implementation experiments, the browser application itself was run on all 15 dataset videos. The application (Vite development server, Chromium) decoded each file with its input limits (480×270, 90 frames, 15 fps), ran the Benchmark Lab suite with the **1-trial (fast check)** setting and default parameters, and its "Export JSON" output was saved unchanged (`results/app_benchmark/*_benchmark.json`; condensed in `app_summary.csv`). No ground-truth masks were loaded, so the application marks IoU, precision, recall and F1 as N/A. Timings are a single trial on a shared workstation and are indicative only.

Table 12.8. Application-reported Jitter RMS (px) of the smoothed trajectory and reduction versus A0; RANSAC inlier ratio; ECC convergence.

Video	A0 jitter	A1 LK+RANSAC	A2 ECC	Inliers (%)	ECC conv. (%)	ECC status
bg_01	0.09	0.011 (88.5 %)	0.007 (93.1 %)	88.4	100.0	OK
bg_02	0.06	0.007 (87.6 %)	0.004 (92.9 %)	96.3	100.0	OK
bg_03	0.12	0.010 (91.8 %)	0.006 (94.9 %)	94.5	100.0	OK
bg_04	0.14	0.025 (82.4 %)	0.025 (82.5 %)	95.3	100.0	OK
bg_05	0.36	0.039 (89.3 %)	0.039 (89.2 %)	96.3	100.0	OK
comb_01	6.15	0.368 (94.0 %)	1.444 (76.5 %)	24.0	88.1	Partial
comb_02	0.44	0.056 (87.2 %)	0.051 (88.4 %)	91.8	100.0	OK
comb_03	0.74	0.106 (85.6 %)	0.118 (84.1 %)	87.0	100.0	OK
comb_04	0.56	0.108 (80.8 %)	0.119 (78.9 %)	85.8	100.0	OK
comb_05	0.94	0.090 (90.4 %)	0.236 (74.8 %)	67.8	100.0	OK
stab_01	1.46	0.049 (96.6 %)	0.273 (81.4 %)	71.5	94.4	Partial
stab_02	0.59	0.024 (95.9 %)	0.095 (83.8 %)	87.2	96.6	Partial
stab_03	7.28	0.483 (93.4 %)	0.304 (95.8 %)	21.8	100.0	OK
stab_04	0.45	0.014 (96.9 %)	0.181 (59.5 %)	72.8	95.5	Partial
stab_05	1.21	0.048 (96.0 %)	0.167 (86.2 %)	48.5	98.8	Partial

The application reports that A1 reduces its own Jitter RMS by 80.8 to 96.9 % (median 90.4 %) and A2 by 59.5 to 95.8 % (median 84.1 %); A1 is ahead of A2 on 9 of 15 videos. A2's ECC solver reported "Partial Fallback" on 5 videos (comb_01, stab_01, stab_02, stab_04, stab_05), meaning that some frames did not converge. The correlation between the inlier ratio and the A1 reduction is r = -0.51 (n = 15), the opposite sign of the +0.39 found with the re-measured jitter in Section 12.3: in the application's own metric the clips with the lowest inlier ratios (comb_01, stab_03) also have the largest starting jitter, so a large

relative reduction is easy to obtain. The application's Jitter RMS therefore cannot confirm or refute the inlier-ratio warning.

Why these numbers differ from Sections 12.3–12.5. The application's Jitter RMS is computed on the trajectory that the same estimator produced, so it measures how well the smoothing removed the estimated shake, not whether the estimate was right. This is the trajectory-based proxy discussed in Section 12.3. The reference re-measurement estimates motion again on the stabilized frames and found that A1 removed only 25 % and 4 % of the jitter on comb_01 and comb_05, whereas the application's own metric reports 94.0 % and 90.4 % for the same clips. The two sets of numbers answer different questions and should not be pooled. In addition, the application's absolute jitter values are smaller than the reference ones (for example stab_01: 1.46 px versus 3.58 px) which most likely reflects its own TypeScript LK/RANSAC code (see the engine notes in the export) and a different trajectory convention; this was not isolated experimentally.

Table 12.9. Median processing time per frame (ms) across the 15 videos: browser application (1 trial) versus the reference implementation (3-trial median).

ID	Algorithm	Application (ms/frame)	Reference (ms/frame)
A1	LK + RANSAC	14.64	6.60
A2	ECC (Euclidean)	71.11	9.10
B1	Frame Differencing	26.94	0.07
B2	MOG2	30.38	2.87
B3	Sample-based KNN	32.86	15.21
C1	Combined (LK+RANSAC → MOG2)	53.71	—

In the browser, A2 is about 4.9 times slower than A1 and Frame Differencing is no longer far faster than MOG2 (26.9 versus 30.4 ms/frame); the likely cause is that the application runs in a browser (JavaScript/WebAssembly with canvas read-backs) whereas the reference is native code; this was not profiled. The ranking A1 < A2 (cost) and the finding that KNN is the slowest background model hold in both settings, but the absolute values should be taken from the application.

The background-model behavior agrees with the reference. On the clouds time-lapse (bg_05), Frame Differencing and KNN label 18.0 % and 19.0 % of the frame as foreground, against 0.7 % for MOG2. Stabilizing before MOG2 (C1) changes the mask flicker by +0.00, -0.90, -0.89, -0.48, -0.52 points on comb_01 to comb_05; it decreases on 4 of 5 combined videos (mean -0.56 points), matching the small, proxy-level benefit found with the reference implementation. These remain proxy metrics: no IoU or F1 can be reported without ground-truth masks.

13. Error Analysis (Diagnostics & Failure Modes)

13.1. When Stabilization Fails

The dominant failure of A1 is a **low RANSAC inlier ratio**. 4 of the ten shaky videos have an inlier ratio below 55 % (stab_04 45 % (jitter reduction 86 %), stab_05 30 % (jitter reduction 42 %), comb_01 42 % (jitter reduction 25 %), comb_05 50 % (jitter reduction 4 %)). Three of them show little benefit; stab_04 is the exception. Large moving foreground (comb_01, comb_05), strong parallax from a near ground plane (the point-of-view walking clips) and motion blur at high shake amplitude each leave too few points that agree on one global similarity transform. The failure is silent in the smoothed-trajectory metric, which still reports an apparently large reduction, so the application's inlier ratio is an essential companion metric.

ECC fails differently. It never aborts, because the solver returns the best estimate after its iteration budget, but for large per-frame displacement or a repetitive texture it converges to a wrong optimum (stab_03 to stab_05), and its trajectory-based jitter then understates the true residual motion. The ECC convergence rate in the reference run was 100 % on every video because OpenCV reports an error only when the optimizer cannot start; the application's own convergence flag, based on its tolerance test, is the more informative measure and should be inspected in the Benchmark Lab.

13.2. Segmentation Failure Modes

- **Illumination drift and dynamic background:** B1 and B3 label 19 to 20 % of the time-lapse clouds clip as foreground against 0.4 % for B2; a fixed or slowly updated reference cannot follow a changing sky.
- **Camera shake:** on the severely shaking comb_01 every model produces large unstable masks (B2 foreground 27.8 %, flicker 27.3 %); neither stabilization (flicker 25.9 %) nor any background model repairs a failed global-motion estimate.
- **Shadows:** disabling shadow suppression (D6) doubles the foreground ratio; shadows of moving people and cars are not part of the object.
- **Noise fragmentation:** without morphology (D5) the number of components per frame rises from about 58 to more than 1,700.

13.3. Border Loss and Dataset Effects

Border loss is proportional to the amplitude of the correction: it is close to 0 % on the static clips, 1 to 2.5 % on mild shake and up to 7 % on the severe clips (Table 12.2), and it grows quickly with the smoothing radius (Section 12.7). The stabilization clips are point-of-view shots in which part of the body is attached to the camera; the stabilizer treats these pixels as background, so some inlier points are on an object that moves with the camera and the motion estimate is biased toward zero. This is a property of the data, not of the algorithms, and a tripod-free handheld pan would be a cleaner test.

14. Discussion

14.1. Answers to the Research Questions

RQ1. With ground truth, LK + RANSAC and ECC recover small, smooth shake about equally well and remove 82 to 97 % of the injected jitter (more than 90 % at the Medium and High levels). On real, severe shake LK + RANSAC is the more reliable of the two in this study (with the caveat that the re-measurement uses the same estimator family); cost is discussed in Section 12.3. When moving foreground or parallax dominates, neither is reliable.

RQ2. On fixed-camera video MOG2 is the most stable model; Frame Differencing is the fastest but cannot follow illumination drift, and the project Sample-based KNN behaves like Frame Differencing under default parameters while being the slowest. Because the real videos have no masks, these statements rest on proxy metrics and the known content of the scenes, not on IoU.

RQ3. Stabilizing first lowered flicker on four of five combined videos by a small amount, which is consistent with but does not prove the benefit of the recommended order. In the ablation the stages that matter most for the mask are shadow suppression and morphology; stabilization-specific stages matter when shake is severe and tracking succeeds (RANSAC doubles the residual jitter when removed).

14.2. Status of the Hypotheses

Table 14.1. Evidence status of the research hypotheses.

Hypothesis	Status	Evidence and caveat
H1 Smoothing reduces trajectory oscillation	Supported	Controlled data: about 3.7 → 0.1–0.2 px at High shake; real mild shake −86 to −89 %. Trajectory metric alone is not proof; see H2 caveat.
H2 LK+RANSAC and ECC differ by motion type	Supported	Equal on smooth synthetic shake; ECC increases jitter on stab_03 to stab_05; re-measurement favors LK (same estimator family).
H3 MOG2 adapts better than Frame Differencing	Supported by proxy only	Time-lapse clouds: 0.4 % vs 19.4 % foreground; no masks, so accuracy is not confirmed.
H4 Stabilize first improves foreground quality	Inconclusive	Flicker −0.3 to −1.4 points on four of five videos; proxy metrics, differences about one point.
H5 Removing stages changes quality and cost	Supported	D5 and D6 change masks most; D2 doubles jitter; D1 lowers proxies unexpectedly.
H6 Processing order matters	Not tested	Only C1 (original → BG) versus C2 (stabilized → BG) were run; C3 (BG → mask alignment) was not implemented in the reference.
H7 Highest quality is not highest performance	Partly supported	Frame Differencing is the fastest but not the most stable under drift; for stabilization the quality and speed comparison is in Section 12.3.

14.3. Practical Recommendations

- Use A1 (LK + RANSAC) as the default stabilizer, check the RANSAC inlier ratio, and treat a low ratio (below about 50 % in this study) as a warning that stabilization may be unreliable.
- Keep the smoothing radius between 10 and 15; larger values trade little jitter for much border loss.
- Use MOG2 with shadow detection and the full morphology chain; the other two models are useful as comparison baselines.
- Report foreground ratio and flicker only as diagnostics; use ground-truth masks (synthetic scenarios or CDnet 2014) for any accuracy claim.

15. Limitations

- **Two measurement sources.** Sections 12.1–12.7 come from a Python/OpenCV re-implementation, and Section 12.8 from the browser application. Their jitter definitions, ECC/RANSAC/KNN details and timings differ and are not pooled. The application was run with the 1-trial setting only, on a development server, on a shared machine.
- **No ground truth on real video.** IoU, F1 and Transform RMSE exist only for the controlled data; real-video segmentation is judged by proxy metrics.
- **Re-measurement bias.** The residual-jitter check uses LK + RANSAC and may favor A1.
- **Small dataset.** Fifteen clips of 4 to 10 s from stock footage; stabilization clips are point-of-view walking shots with parallax and camera-attached body parts; three of the five combined clips have only mild shake.
- **Statistics.** The quality metrics are deterministic and were computed once per configuration; only timing uses a three-trial median. The paired tests, block bootstrap and confidence intervals provided by the application were not run, and no p-values are claimed.
- **2D models.** A global similarity or Euclidean transform cannot remove parallax (Liu et al., 2013); residual error on the walking clips is partly of this kind.
- **Scope of experiments.** Processing order C3 and the theoretical C4 were not evaluated; CDnet 2014 was not used because it could not be downloaded in the working environment.

16. Ethics Statement

All videos are third-party stock footage released under the Pexels License and were used unaltered in content (only trimmed, scaled and re-encoded). Several clips show people in public places. The study performs motion-based segmentation only; no face, identity or behavior recognition is attempted and no personal data were collected, so informed consent was not required. The dataset is shared for coursework and should not be redistributed outside the terms of the Pexels License. No experimental result, dataset or reference was fabricated: every number is produced by the scripts in Section 18, and the limits of the evidence are stated in Section 15.

17. Conclusion

This report documented the SpatioTemporal Video Processing Lab, assembled a licensed 15-video dataset in three categories and ran the project's experiment groups with a reference implementation that follows the application's defaults and with the application's own Benchmark Lab. With known camera motion, both stabilizers remove more than 90 % of the injected shake at the Medium and High levels (High level: 3.68 px before, about 0.14 px after). On real video the picture is more nuanced: LK + RANSAC works when most tracked points follow one global motion and fails under heavy foreground or parallax, ECC is less robust to large displacement, and the trajectory-based proxy can overstate success. For background subtraction, MOG2 is the most stable model under illumination drift, and shadow suppression and morphology have the largest effect on mask quality. Stabilizing before background subtraction gave only small, proxy-level improvements. The central open item is quantitative segmentation accuracy, which needs ground-truth masks; the next steps are to evaluate Scenario 1 and CDnet 2014 camera-jitter sequences with IoU and F1, and to repeat the application runs with the three-trial protocol and paired statistics.

18. Reproducibility and Source Code

- **Source code:** https://github.com/bnvtoi5/videoprocessor (React 19, TypeScript, Vite, OpenCV.js 5.0.0-release.1).
- **Dataset (15 videos, metadata.csv, README):** https://drive.google.com/drive/folders/1jula3nmZzVjUM8j76i8jq4CKwTAkWs7k
- **Run the application:** `npm install`, then `npm run dev` (port 3000); unit tests: `npx --yes tsx --test tests/algorithms.test.ts`; type check: `npm run lint`. Upload a dataset video in the Dashboard (defaults: 90 frames, 480 px, 15 fps) and use the Stabilization, Background, Combined and Benchmark Labs.
- **Reference experiments (this report):** Python 3.10, OpenCV 4.13 (`opencv-python-headless`), NumPy, Matplotlib. Scripts: `exp.py` (Modules A–C, ablation, sensitivity), `synth.py` (controlled validation with known camera motion), `make_figs.py` (figures), `build_report.py` (this document). All parameters are the application's defaults and the synthetic shake uses a fixed random seed (11), so the outputs are deterministic apart from timings.
- **Note:** ground-truth metrics are shown as "N/A – Ground Truth Unavailable" in the application for the real videos by design.

19. References

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Note: bibliographic metadata was taken from the project's `literatureDatabase.ts`, from the CVF/HAL records of the CDnet 2014 paper and from the Pexels license page; the group should re-verify each DOI before submission.

Appendix A – Literature Matrix

Table A.1. Summary of the principal works used in this study.

Study	Method / contribution	Role in this project	Limitation relevant here
Lucas & Kanade, 1981	Gradient-based iterative image registration (optical flow)	Feature tracking core of A1	Assumes small displacement and local brightness constancy
Shi & Tomasi, 1994	Minimum-eigenvalue corner selection	Corner detector in A1 (`maxCorners`, `qualityLevel`)	Corners also fall on moving objects
Fischler & Bolles, 1981	RANSAC robust model fitting	Outlier rejection in A1; ablation D2	Needs a sensible inlier threshold
Evangelidis & Psarakis, 2008	ECC maximization, illumination-invariant alignment	Method A2	Heavier than sparse tracking; may not converge for large displacement
Grundmann et al., 2011	Optimal L1 camera paths	Motivates trajectory smoothing and Jitter RMS	Stronger smoothing increases crop/border loss
Liu et al., 2013	Bundled (mesh) camera paths	Explains residual parallax error of global 2D models	Not implemented here
Zhang et al., 2019	Motion-stability assessment	Justifies separating proxy and ground-truth metrics	Smoothness alone is not quality
Piccardi, 2004	Review of background subtraction techniques	Taxonomy for B1/B2/B3	Review, no experiments here
Stauffer & Grimson, 1999; Zivkovic, 2004	Adaptive Gaussian mixtures; adaptive component count	Method B2 (MOG2)	Pixel-wise stationarity broken by camera shake
Zivkovic & van der Heijden, 2006; Barnich & Van Droogenbroeck, 2011	Non-parametric KNN density; ViBe sample bank	Method B3 (sample-based KNN)	Cost grows with sample count
Chapel & Bouwmans, 2021	Review of moving-object detection with moving cameras	Context for combined pipeline	Survey
Wang et al., 2014	CDnet 2014 benchmark (camera-jitter category)	Recommended ground-truth benchmark for future work	Not used here (not downloadable in this environment)

Appendix B – AI Declaration

Table B.1. AI usage declaration.

Item	Description
AI Tool	Claude (Anthropic), used through the Claude Code command-line agent.

Item	Description
Purpose	Searching and selecting candidate videos on Pexels; downloading, trimming and re-encoding them; measuring their camera jitter and motion statistics; reading the project source code and documentation; writing a Python/OpenCV reference re-implementation and the experiment scripts; installing the application and running its Benchmark Lab on every video; producing the figures; drafting the text of this report in the DSP501 format.
Stage Used	Dataset construction, experiment execution, result analysis, figure generation and report drafting.
Human Verification	To be completed by the group before submission: review every statement against the source code, re-run the scripts (Section 18), confirm the numbers, and replace the placeholders on the title page.
Final Responsibility Statement	The group remains responsible for the originality, correctness and integrity of the submitted work. No experimental number in Section 12 was typed by hand: every value is read from the JSON output of the scripts listed in Section 18 or from the Benchmark Lab export of the application (Section 12.8). Where evidence is a proxy or a different implementation than the browser application, the report says so.

Appendix C – Project Checklist

Table C.1. Deliverable status at the time of writing.

Project Deliverable / Milestone Item	Verification Status
Research problem, objectives, questions, hypotheses	[x]
Literature review and research gap	[x]
DSP pipeline documented from source code	[x]
Experimental design and ablation matrix	[x]
Baseline comparison (A0, Flow A)	[x]
Ablation study (reference implementation)	[x]
Parameter sensitivity (reference implementation)	[x]
Error analysis and discussion	[x]
Ethics statement	[x]
AI declaration (group verification pending)	[x]
Final report (this document)	[x]
Source code on GitHub	[x]
Dataset on Google Drive (15 videos + metadata)	[x]
Benchmark Lab export from the browser application (1 trial, 15 videos)	[x]

Project Deliverable / Milestone Item	Verification Status
Individual AI reflection	[ ]
Demo video	[ ]
Presentation	[ ]

Appendix D – Video Sources and Licenses

Table D.1. Source, author and license of every dataset video.

Video	Author (Pexels)	Source page
stab_01_pov_leaves	K (Pexels @kelly)	pexels.com/video/a-person-walking-outdoors-5597043/
stab_02_pov_wet_pavement	K (Pexels @kelly)	pexels.com/video/a-person-holding-a-camera-while-looking-down-and-walking-4310191/
stab_03_pov_brick_pavement	Athena Sandrini	pexels.com/video/person-walking-3002456/
stab_04_pov_sidewalk	K (Pexels @kelly)	pexels.com/video/a-person-taking-video-while-walking-4405576/
stab_05_pov_dirt_path	K (Pexels @kelly)	pexels.com/video/point-of-view-of-a-person-walking-3978362/
bg_01_intersection_traffic	German Korb	pexels.com/video/traffic-flow-in-an-intersection-4791734/
bg_02_winter_street	German Korb	pexels.com/video/fixed-shot-of-an-intersection-in-montreal-3785878/
bg_03_london_street	George Morina	pexels.com/video/vehicles-stopping-at-the-traffic-light-5330908/
bg_04_traffic_cameras_bus	Grigoriy Bunkov	pexels.com/video/busy-urban-street-monitoring-with-traffic-cameras-36553218/
bg_05_timelapse_clouds	Zuzanna Musial	pexels.com/video/cloudy-day-and-a-busy-road-1388383/
comb_01_shaky_people	WeStarMoney Rec	pexels.com/video/shaky-footage-of-people-walking-on-the-road-side-4101528/
comb_02_aerial_intersection	michael hernandez	pexels.com/video/intersection-19130216/
comb_03_night_crowd	Ludovic Delot	pexels.com/video/vibrant-nighttime-street-celebration-with-crowd-38471369/
comb_04_crowd_crossing	Alex Pandian	pexels.com/video/people-walking-at-a-commercial-district-4583056/
comb_05_runner_street	Alin Serban	pexels.com/video/scenic-urban-morning-run-in-small-european-town-37283732/

License for all videos: Pexels License (https://www.pexels.com/license/). Videos were trimmed to at most 10 s, scaled to at most 640 px width and re-encoded to H.264 without audio; no other modification was made.